CEOVIA

GLOBAL DIGITAL BRAND PLATFORM

Technical Specification & Website Blueprint

Production-Ready Architecture for Category-Leading Performance

Version	3.0 — Unified & Refined
Date	April 2026
Status	Production-Ready Architecture
Target	Top 1% performance, conversion, and compliance in beauty/wellness

EXECUTIVE VISION

Platform Thesis
CEOVIA will operate as an Agentic Commerce Platform — where AI agents, social discovery, and headless commerce converge to deliver 40%+ conversion lifts over traditional ecommerce. Core differentiators: Sub-1.2s LCP globally via Next.js 15 Partial Prerendering · Real-time social commerce with sub-5s inventory sync · Zero-trust compliance — PII masked at edge, claims validated pre-publish · AI-first discovery — optimised for ChatGPT, Perplexity, and Gemini citations.

THE 8-LAYER ARCHITECTURE — OVERVIEW

#	Layer	Technology	Performance Target
1	Edge Security & Compliance	Cloudflare Workers + DLP	100% PII masked pre-processing
2	Experience Layer	Next.js 15 (PPR) + Vercel Edge	<1.2s LCP, 100% Lighthouse
3	Commerce Engine	Shopify Plus + Hydrogen/Oxygen	99.99% uptime, 40K checkout/min
4	Content Platform	Sanity (structured, localised)	<50ms query response
5	Social Commerce	TikTok Shop + Instagram Graph API	<5s inventory sync, real-time UGC
6	AI & Intelligence	OpenAI Responses API + RAG	4× conversion rate
7	Search & Discovery	Algolia + AEO schema	90%+ AI citation accuracy
8	Governance & Data	PostgreSQL 16 + Redis 7	Zero compliance incidents

LAYER 1: EDGE SECURITY & COMPLIANCE

Paradigm Shift
Security and privacy are no longer middleware — they are the entry point for every user request. All traffic passes through Cloudflare Edge before reaching any application layer.

1.1 Cloudflare Workers Architecture

User Request → Cloudflare Edge (275+ PoPs) → WAF + Bot Management → PII Detection & Masking (DLP) → Geographic Routing (GDPR/HIPAA zones) → Sanitised Request → Vercel

1.2 PII Detection — Health Pattern Rules

Pattern Type	Example Trigger	Action
Health Condition	"acne history photos", "eczema severity"	Mask — remove before AI processing
Treatment Reference	"laser treatment session", "isotretinoin mg"	Mask — anonymise and log detection
Before/After Imagery	Image metadata: "before-after", "treatment-progress"	Quarantine — human review required

1.3 Compliance Zones

Region	Data Residency	Regulation	Special Handling
US	Cloudflare US East/West	HIPAA / DSHEA	BAA required for health data
EU	Cloudflare EU Central	GDPR / EFSA	Data processing agreement; right to erasure
GCC	Cloudflare ME	Saudi FDA / UAE MOH	Data localisation; Arabic content compliance
APAC	Cloudflare Singapore	PDPA	Consent-first data capture

LAYER 2: EXPERIENCE LAYER (NEXT.JS 15 + PPR)

2.1 Architecture Pattern

Partial Prerendering (PPR) delivers static shells instantly, then streams dynamic content. This achieves sub-1.2s LCP by serving HTML from CDN edge while personalisation loads asynchronously.

Content Layer	Render Strategy	Performance Target
Layer 1: Static shell	CDN edge pre-render	<100ms — product images, brand assets, page structure
Layer 2: Dynamic content	Streamed (Suspense)	<500ms — pricing, inventory, reviews
Layer 3: Personalisation	Lazy (async)	<1s — AI recommendations, user-specific content

2.2 Performance Targets vs Industry

Metric	CEOVIA Target	Industry Average	Implementation
LCP (mobile)	<1.2s	2.5–4s	PPR + Edge caching
TTFB	<50ms	200–600ms	Cloudflare + Vercel Edge
FID / INP	<50ms	100–300ms	React 19 concurrent features
CLS	<0.05	0.1–0.25	Strict layout enforcement

LAYER 3: COMMERCE ENGINE (SHOPIFY PLUS)

3.1 Multi-Store Architecture for Global Scale

Store	Domain	Market	Currency	Fulfilment	B2B
Primary	ceovia.com	US	USD	US 3PL	Shopify B2B
UK	ceovia.co.uk	United Kingdom	GBP	UK 3PL	Expansion store
EU	ceovia.eu	EU (GDPR)	EUR	NL Hub	Expansion store
GCC	ceovia.ae	UAE / Saudi	AED/SAR	Dubai Hub	Expansion store
Pro (B2B)	pro.ceovia.com	Global B2B	Multi	Direct	Dedicated B2B

3.2 Checkout Architecture

- Shopify Checkout — highest conversion; do not replace
- UI Extensions for customisation (compliance banners, market-specific disclaimer injection)
- Recharge for advanced subscriptions, bundling, and loyalty
- Shopify B2B for clinic and distributor wholesale portal at pro.ceovia.com

LAYER 4: CONTENT PLATFORM (SANITY)

4.1 Content Model for AEO (Answer Engine Optimisation)

AEO Imperative
55%+ of Google searches now return AI Overviews. CEOVIA must be structured for AI citation, not just keyword ranking. This requires entity-first content architecture where ingredients, studies, claims, and products are stored as structured data objects — not plain text.

Content Entities (Sanity Schema Objects)

Entity	Key Fields	AI/SEO Purpose
Ingredient	scientificName, inciName, mechanismOfAction (max 3), clinicalStudies, approvedClaims	Enables AI citation of mechanism; grounds RAG responses
Clinical Study	pubmedId, studyType (RCT/in-vitro/meta), participantCount, results (max 200 chars), doi	Powers Evidence Badges; enables Gold/Silver/Bronze classification
Approved Claim	text, category, status, sources, applicableMarkets	Market-validated claims; prevents unapproved Layer 3 → Layer 1 migration
Blog Post	title, occasionTags, layerClassification, evidenceBadges, disclaimer, aiSearchBlocks	Occasion tagging for AISO; AEO-structured FAQ blocks
Market Rule	market_code, currency, allowed_claims, socialRestrictions, influencerDisclosureReqs	Drives market-specific claim filtering in Compliance Critic Agent

4.2 Layer Classification in CMS

Enforcement Rule
Every content item in Sanity must be tagged with a Layer (1, 2, or 3). The publishing workflow must prevent Layer 2 or Layer 3 content from being routed to Layer 1 pages (Homepage, product pages, ads, social). This is enforced in the CMS validation schema, not just in guidelines.

LAYER 5: SOCIAL COMMERCE (TIKTOK SHOP + INSTAGRAM)

5.1 Strategic Position

TikTok Shop generated over $33B in GMV in 2024. Instagram Shopping tags drive 42% sales increases. Social commerce is the primary discovery channel for CEOVIA's target demographics — not an optional add-on.

5.2 TikTok Shop Integration

Function	API Endpoint	Implementation	Data Flow
Product Sync	POST /product/create	Shopify → TikTok Shop	Bidirectional via middleware
Inventory Sync	POST /inventory/update	Real-time webhook	ERP → TikTok Shop (<5s)
Order Retrieval	GET /order/list	Polling + webhooks	TikTok Shop → Shopify
Creator Affiliate	Partner API	Campaign management	Internal → TikTok

5.3 Viral Moment Protection

- Redis queue buffers 10,000+ events per minute
- Circuit breaker: If Shopify API fails, pause TikTok sales — never oversell
- Tiered inventory allocation: reserve 20% buffer for D2C; 80% for TikTok during viral spikes
- Real-time dashboard: Inventory sync latency <5s target

5.4 Instagram Shopping

- Automated product feed sync from Shopify to Meta Commerce Manager
- Product tagging in posts, Stories, and Reels via Instagram Graph API
- Shoppable content embedding on CEOVIA website
- Instagram Checkout integration where available by market

LAYER 6: AI & INTELLIGENCE (OPENAI + RAG)

Governance Rule
Architecture = Retrieval-Augmented Generation with strict governance. The AI never generates unapproved claims. Every response is grounded in the verified CEOVIA claims database. Open-model generation of citations is prohibited — see AI Agent Ecosystem document, Part 4.2 for Reasoning Snippet governance rules.

6.1 AI Governance Context Object

Context Field	Source	Purpose
userMarket	Cloudflare geo-detection	Load market-specific approved claims
userType	Progressive profiling / HubSpot	Adjust explanation depth (consumer vs. clinic)
approvedClaims	PostgreSQL claims registry	Only verified claims; never generated
ingredientScience	Sanity ingredient entities	Grounds mechanism-of-action responses
conversationHistory	Redis (90-day memory)	Stateful 90-day protocol support
socialProof	Social aggregation service	Supplementary only — labelled "community favourite," never clinical advice

6.2 Response Validation (4-Layer Process)

Step	Name	Action
1	Hard claim validation	Extract implied claims from user message. Block if not in approvedClaims database.
2	Grounded prompt build	Assemble prompt using ONLY context from approved database — no open generation.
3	Governed generation	OpenAI Responses API with temperature 0.1 (low creativity) and max 500 tokens.
4	Post-generation validation	Re-validate output against approvedClaims. Log and block if violation detected. Use safe fallback.

LAYER 7: SEARCH & DISCOVERY (ALGOLIA + AEO)

7.1 AI Search Optimisation (AISO) — Occasion Tagging

Traditional optimisation targets keywords. CEOVIA targets life scenarios. This is how content ranks in ChatGPT, Perplexity, and Gemini — by matching life occasions, not product features.

Occasion Tag	Trigger Keywords	Target AI Query
Skin hydration during air travel	"jet lag", "dry skin flight", "travel fatigue"	"What to take for dry skin on long flights"
High-stress executive recovery	"burnout", "executive performance", "fatigue"	"Best supplement for burnout recovery"
Pre-event skin prep	"wedding skin", "glow before event", "photoshoot"	"How to get glowing skin before a big event"
Post-antibiotic microbiome support	"after antibiotics", "gut restore", "microbiome"	"What to take after antibiotics to restore gut"
Morning ritual sustained focus	"morning energy", "coffee alternative", "clarity"	"Alternative to coffee for all-day energy"

Compliance Note
All occasion tags must use Layer 1 language. "Supports," "helps maintain," "designed to support" — never "treats," "prevents," or disease names. Occasion tagging is a marketing and SEO function, not a clinical claim framework.

LAYER 8: GOVERNANCE & DATA

8.1 Claims Governance System

Data Model	Key Fields	Purpose
Claim	id, text, category, status, sources, applicableMarkets	Single source of truth for all approved claims
SocialContent	platform, externalId, creator, products, approvedForSite, rightsStatus, marketVisibility	Controls which UGC can appear on CEOVIA properties
MarketRule	market_code, currency, allowed_claims, socialPlatformRestrictions, influencerDisclosureReqs	Drives Compliance Critic market-specific filters
CreatorPartnership	creatorId, contractStatus, approvedMarkets, commissionStructure, disclosureTerms	FTC / global compliance for influencer content
AIInteraction	userId (hashed), query, response, claimsUsed, violations, timestamp	7-year audit trail for regulatory review

8.2 Non-Negotiable Data Rules

#	Rule	Enforcement
1	AI never generates unapproved claims	Middleware claim validation before every response
2	Product truth lives in structured data (Sanity)	CMS validation schema; no plain-text claims
3	No market-specific website forks	Single codebase; market rules in database
4	All AI interactions logged	Accountability; 7-year retention
5	Sub-3-second LCP globally	Edge rendering; PPR; Cloudflare CDN
6	All social content requires brand safety review	approvedForSite field must be true before display
7	Creator partnerships require contract tracking	CreatorPartnership data model with contractStatus field
8	Social platform inventory sync <60 seconds	Real-time webhooks + Redis buffer + circuit breaker
9	User data erasure capability must be built before launch	Redis erasure endpoint + privacy policy documentation
10	Schema, localisation, and compliance are launch requirements — not phase 2	Pre-launch checklist; do not defer

WEBSITE STRUCTURE & PAGE ARCHITECTURE

9.1 Navigation Architecture (Mandatory)

B2B Navigation Rule
Clinics and distributors are core revenue drivers — not secondary audiences. Top-level navigation must explicitly surface B2B pathways. AI routing supplements navigation; it does not replace it.

Nav Item	Audience	Key Pages
For Consumers	B2C	Homepage → Products → 90-Day System → Science → Blog → FAQ
For Clinics	B2B	/for-clinics → Practitioner Portal (pro.ceovia.com) → Clinical Summary Download
For Distributors	B2B	/for-distributors → Partnership Application → Volume Pricing Guide
Science	Layer 2/3	/science → Ingredients → Clinical Evidence → Study Citations → Evidence Badges
Community / Social	All	/community (Social hub) → /social-shop (Shoppable videos) → /creators (Creator directory)

9.2 Page Content Layer Rules

Page	Content Layer	Rule
Homepage	**Layer 1 ONLY**	No clinical terms. No condition names. "Supports," "helps maintain" language only.
Product Pages	**Layer 1 ONLY**	Benefits in lifestyle language. Clinical mechanism links to Science page, not inline.
Blog / Content	**Layer 1–2**	Educational depth permitted. Evidence Badges required. Occasion-tagged. Mandatory disclaimer.
Science Page	**Layer 2–3**	Mechanism-of-action, study citations, Evidence Badges. Full context and limitation notes.
Practitioner Portal	**Layer 2–3**	Gated. Full clinical language. Prescribing notes. Contraindication details.
Social / Community	**Layer 1**	All claims Layer 1 compliant. UGC must pass brand safety review (approvedForSite = true).

IMPLEMENTATION ROADMAP

Phase 1: Core Platform (Months 1–3)

- Next.js 15 + Vercel + Tailwind setup
- Shopify Plus integration (products, checkout, subscriptions, Recharge)
- Sanity CMS (content models, Layer classification, editorial workflows)
- Cloudflare security (WAF, DLP, PII masking, geo-routing)
- Basic SEO / structured schema
- Meta Commerce Manager setup and Instagram Shopping approval
- User consent architecture and data erasure endpoint

Phase 2: Search, Social & Discovery (Months 4–5)

- Algolia integration + AEO schema + Occasion Tagging for AISO
- Instagram Shopping product tagging automation
- TikTok Shop seller account setup + basic product sync
- Social content aggregation (manual curation initially)
- Subscription management (Recharge)

Phase 3: Governance & B2B (Months 5–6)

- PostgreSQL governance database (Claims, MarketRule, CreatorPartnership models)
- Claims approval workflow + market-specific configuration
- B2B portal (/for-clinics, /for-distributors, pro.ceovia.com) with Shopify B2B
- Creator partnership tracking
- Automated disclaimer injection per market

Phase 4: AI & Advanced Social (Months 6–8)

- AI middleware with full governance context object
- OpenAI Responses API + RAG integration with verified claims database
- Social context in AI responses (labelled "community favourite" — never clinical)
- Automated TikTok Shop order sync + viral moment protection
- Social hub (/community) launch + shoppable video integration

Phase 5: Answer Engine & Scale (Months 8–10)

- Entity-first content architecture fully deployed in Sanity
- AI-citable answer blocks and FAQ schema
- Creator affiliate programme automation
- Multi-market expansion automation

Phase 6: Personalisation & Community (Months 10–12)

- CRM integration (Segment + Klaviyo)

- Advanced personalisation engine
- Private community platform (Circle or Discord)
- Lifecycle marketing automation + predictive retention

CORE TECHNOLOGY STACK — SUMMARY

Layer	Technology	Purpose	Deployment
Frontend	Next.js 15 (App Router) + TypeScript 5.4	React-based SSG/SSR/ISR/PPR	Vercel Edge
Styling	Tailwind CSS 3.4 + Radix UI + Framer Motion	Utility-first with design tokens	Vercel
Commerce	Shopify Plus + Hydrogen/Oxygen	Product catalog, checkout, subscriptions	Shopify Cloud
CMS	Sanity	Structured content, editorial workflows	Sanity Cloud
Social Commerce	TikTok Shop Open API + Instagram Graph API	Social selling, shoppable content	Multi-platform
Search	Algolia	Product search, AEO, recommendations	Algolia Cloud
AI	OpenAI Responses API + Custom Middleware + RAG	Governed intelligence, concierge	Vercel Edge
Database	PostgreSQL 16	Governance, claims registry, AI logs	Supabase / AWS RDS
Cache / Sessions	Redis 7	Rate limiting, sessions, AI memory	Upstash / Redis Cloud
Edge Security	Cloudflare Workers + WAF + DLP	CDN, security, PII masking	Cloudflare Edge
Analytics	GA4 + Google Search Console + Platform APIs	Behavioural analytics, social ROI	Google Cloud
Subscriptions	Recharge (Shopify app)	Advanced subscriptions, bundling, loyalty	Recharge Platform
Community	Circle or Discord (gated)	Private community, expert access	Circle.so / Discord

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